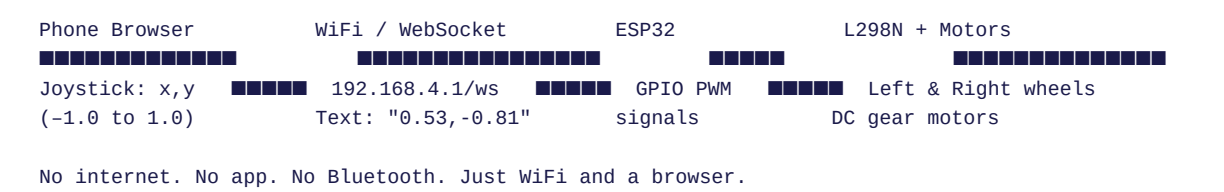


SECTION 1

How It Works

Your joystick position on the phone becomes an (x, y) pair and travels over WiFi as a WebSocket message to the ESP32. The ESP32 mixes those values into left/right motor speeds and drives the L298N motor driver, which supplies enough current to spin two DC gear motors.



The ESP32 also runs a 400 ms fail-safe: if no command is received (WiFi drops, phone locks, tab closes), the motors stop automatically.

SECTION 2

Parts List — What to Buy

All parts are available on Amazon, AliExpress, or Robu.in. Buy exactly what is listed — parts that look similar are often incompatible.

Component	Spec	Notes	Cost (approx)
ESP32 Dev Board (WROOM-32)	ESP32-WROOM-32 38-pin, 3.3V	The brain. Must say WROOM-32 on the metal module.	\$3.00-5.00
L298N Motor Driver	L298N chip 5–35V, 2A/ch	Comes with ENA/ENB jumpers — REMOVE BOTH.	\$5.00-15.00
DC Gear Motors × 2	TT motor, 3–6V ~200 RPM	Yellow plastic gearbox "TT motors". Buy a 4-pack for \$1.00-2.00.	\$1.00-2.00
Wheels × 2	65 mm TT shaft fit	Must fit the D-shaped TT motor shaft. Often sold with motors — check listing.	\$0.60-1.20
18650 Li-Ion Battery × 2 + Dual holder	3.7V each 2× series = 7.4V	Buy name-brand cells (Samsung, LG). Cheap cells sag under motor load.	\$3.00-5.00
2WD Acrylic Chassis	Includes mounts casters, standoffs	Search "2WD Smart Car Chassis Kit". This is the basic one with everything mounts on.	\$1.50-2.50
Dupont Jumper Wires	Male-to-Female 20 cm, 40-pack	M-to-F for ESP32 pins to L298N terminals. Get 40+, \$5-10 cheap.	\$5-10
Micro-USB Data Cable	DATA cable only	For programming the ESP32. Charge-only cables look identical but will NOT work.	\$1.00-2.00
Small flathead screwdriver	Jeweler size	For L298N screw terminals. Finger-tightening causes motors to disconnect mid-drive.	\$1.50

Watch out: kit substitutions

Many cheap kits bundle an L9110S driver (blue board) instead of the L298N. The pinout is different and this code will not work with it. Buy parts separately so you know exactly what you are getting.

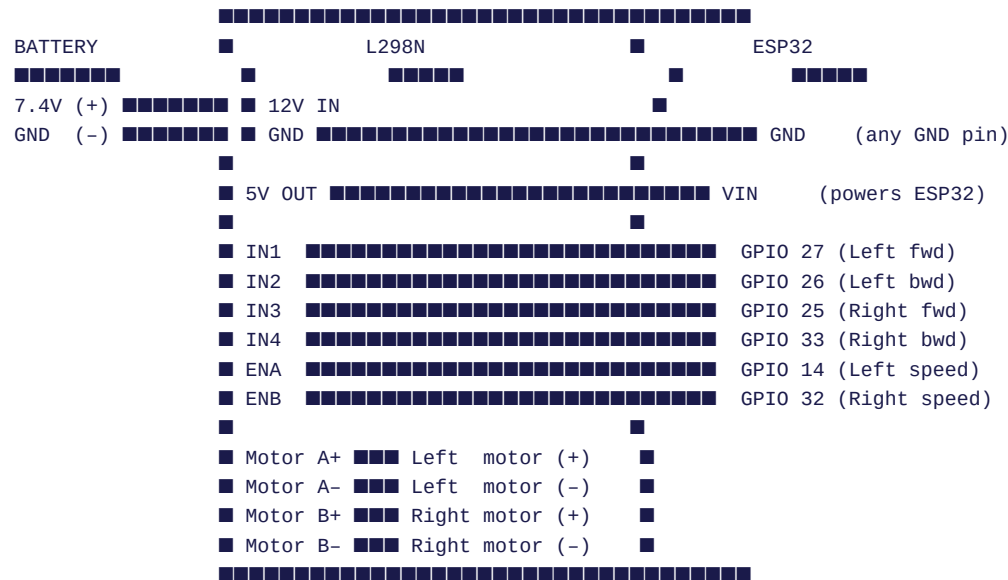
SECTION 3

Wiring

Before you connect anything

- Remove both ENA and ENB jumpers from the L298N board. These are small plastic caps — pull them off with your fingers. Without doing this, the motors only run at full speed.
- Keep the battery disconnected or switched off throughout wiring.
- The ESP32 and L298N MUST share a common GND wire — this is the most commonly missed step.

Connection diagram



Full wiring table

Wire	From	To	Purpose
Battery +	Battery pack (+)	L298N 12V terminal	Motor power in
Battery –	Battery pack (–)	L298N GND terminal	Motor ground
Common GND	ESP32 GND pin	L298N GND terminal	REQUIRED — shared reference
5V supply	L298N 5V OUT	ESP32 VIN pin	Powers ESP32 from battery
Blue	ESP32 GPIO 27	L298N IN1	Left motor forward
Purple	ESP32 GPIO 26	L298N IN2	Left motor backward

Blue	ESP32 GPIO 25	L298N IN3	Right motor forward
Purple	ESP32 GPIO 33	L298N IN4	Right motor backward
Green	ESP32 GPIO 14	L298N ENA	Left motor speed (PWM)
Green	ESP32 GPIO 32	L298N ENB	Right motor speed (PWM)
Red / Black	L298N Motor A+/A-	Left motor terminals	Left motor power
Red / Black	L298N Motor B+/B-	Right motor terminals	Right motor power

Common GND — do not skip this

The common GND wire between ESP32 and L298N is critical. Without it, the ESP32's 3.3V logic signals have no reference point and the motors will not respond — or the ESP32 will behave erratically.

Tip: powering the ESP32 from the battery

When the battery is connected, the L298N's 5V output regulator can power the ESP32 through its VIN pin. During programming, use USB as normal — no wires need to be removed.

SECTION 4

Assembly — Step by Step

Follow this order. Skipping steps or doing them out of order is how components get damaged.

1**Build the chassis**

Screw the TT motors into the motor mounts. Attach wheels and caster balls. The chassis should sit flat and roll freely. Do not overtighten plastic standoffs — acrylic cracks.

2**Remove ENA and ENB jumpers from L298N**

Find the two small plastic jumper caps bridging the ENA and ENB pin pairs. Pull them off with fingers or tweezers. Without this, speed control does not work.

3**Mount boards on chassis**

Use double-sided tape or M3 screws to secure the L298N and ESP32 on the chassis. Keep the ESP32 accessible for USB programming.

4**Connect motors to L298N output terminals**

Loosen the terminal screws, insert motor wires into Motor A (left) and Motor B (right), tighten firmly. A loose motor connection is the most common cause of a car that only steers one way.

5**Route battery wires — do not connect yet**

Run battery (+) to L298N 12V terminal and battery (–) to GND terminal. Leave the battery disconnected or switched off.

6**Connect GPIO control wires (Dupont M-F wires)**

Wire each GPIO pin to the matching L298N input:

- GPIO 27 → IN1
- GPIO 26 → IN2
- GPIO 25 → IN3
- GPIO 33 → IN4
- GPIO 14 → ENA
- GPIO 32 → ENB

7**Connect common GND**

Run a wire from any GND pin on the ESP32 to the GND terminal on the L298N (same terminal as the battery negative). This is the step most people miss.

8 Flash the firmware

Connect ESP32 to PC via USB. Open the .ino file in Arduino IDE 2, select Board "ESP32 Dev Module", select the COM port, click Upload. Open Serial Monitor at 115200 baud — you should see the hotspot IP printed.

9 First test — USB power, motors off

With the battery still disconnected, power via USB. Connect your phone to "ESP32-RC-Car" WiFi (password: rccar123). Open 192.168.4.1. Move the joystick and check the Serial Monitor for received x,y values. This confirms the WiFi/WebSocket pipeline before adding motor power.

10 Full test with battery

Unplug USB. Switch on the battery. The ESP32 now draws power from the L298N 5V output. Connect your phone and drive. If a motor runs backward, swap its two wires at the L298N terminal — never change the code for this.

SECTION 5

Arduino IDE Setup

1 Download Arduino IDE 2

Go to arduino.cc/en/software. Download "Arduino IDE 2" (not the 1.8.x Windows Store version — it has a broken downloader for large packages).

2 Add the ESP32 boards URL

File → Preferences → Additional Boards Manager URLs. Paste:
https://dl.espressif.com/dl/package_esp32_index.json

3 Install ESP32 board package

Tools → Board → Boards Manager → search "esp32" → install "esp32 by Espressif Systems" version 3.x or later.

4 Install two required libraries

Tools → Manage Libraries. Search and install both:

- ESPAsyncWebServer (author must be "ESP32Async")
- AsyncTCP (same author "ESP32Async")
- If not found, download the ZIP from GitHub and use Sketch → Include Library → Add .ZIP

5

Select board and port, then upload

Tools → Board → ESP32 Arduino → "ESP32 Dev Module". Tools → Port → select your COM port. Click Upload. Some boards need you to hold BOOT while upload starts — release once you see "Connecting...".

SECTION 6

Using the Controller

Connect your phone or laptop to the "ESP32-RC-Car" WiFi network (password: rccar123), then open a browser to 192.168.4.1 or esp32car.local.

Mode	How to use	Notes
Joystick (default)	Drag the circle in any direction	Drag angle = turn, drag distance = speed
Arrow keys	Use keyboard arrow keys	Useful for desktop testing
STOP button	Tap the red STOP button	Instantly stops both motors
Reverse Mode	Hamburger menu (top right) → toggle	Flips all directions. Red banner shows when active.
Rainbow Mode	Hamburger menu → toggle	Changes control button colours. Cosmetic only.
Steering Wheel	Hamburger menu → toggle	Rotate the wheel to steer. Hold Gas / Brake pedals.

SECTION 7

Professional Advice

These are things that take beginners hours to debug and experienced builders do on instinct.

Test signal path before motor power

Test USB + WiFi first, then add motor power. If something breaks, you will know exactly which addition caused it.

Motor direction is not predictable

Which way a motor spins depends on which wire goes to which terminal. The fix is always to swap the two motor wires at the L298N terminal, never to change code. Doing it in code will confuse you later.

The L298N heatsink gets warm — that is normal

The L298N burns off about 30% of input power as heat. Warm is fine. Too hot to touch after 30 seconds means your motor voltage is too high or the motors are stalling.

WiFi range is typically 10–20 metres

Concrete walls, metal chassis parts, and motor noise reduce range. Keep the ESP32 antenna end pointing toward you, not at the ground.

Battery quality matters more than capacity

A cheap 18650 cell sags to 6V under motor load, making the car sluggish. A quality cell (Samsung, LG) maintains 7V+. This is the single biggest performance upgrade for under £200.

The 400 ms fail-safe is intentional

If WiFi drops or the phone locks, the car stops within 400 ms. Do not increase this value. A car that keeps running when you lose control will hit something.

For demos: show fun modes last

Lead with basic joystick — intuitive and immediately impressive. Then show Reverse Mode. Save Steering Wheel Mode for last — it is the most technically interesting and ends on a high note.

SECTION 8

Troubleshooting

Page will not load / connection overlay stays up

- Check your phone is on ESP32-RC-Car, not your home WiFi
- Try the IP directly: 192.168.4.1
- Open Serial Monitor — does it show "Server started"? If not, reflash.
- Check the USB cable is a data cable, not charge-only

Motors do not move at all

- Check the common GND wire between ESP32 and L298N is connected
- Confirm both ENA/ENB jumpers have been removed from the L298N
- Verify the battery is on and connected to the 12V terminal
- Check battery voltage with a multimeter — should be 7V or above

Only one motor moves

- Check the non-moving motor wires are firmly in the L298N terminal
- Swap the non-moving motor to the working terminal to test the motor itself
- Verify IN3/IN4 and ENB wiring matches the GPIO numbers exactly

Car moves but steering is wrong

- Swap the backward motor's two wires at the L298N terminal
- If the whole car goes backward on forward, use Reverse Mode in the UI

Board crashes or reboots in a loop

- Ensure you are on Arduino-ESP32 core v3.x, not 2.x
- Check that no boot-strapping pin (0, 2, 5, 12, 15) is being driven externally at boot

SECTION 9

Quick Reference

GPIO pin assignments

GPIO 27 → IN1	Left motor forward	GPIO 14 → ENA	Left motor speed (PWM)
GPIO 26 → IN2	Left motor backward	GPIO 32 → ENB	Right motor speed (PWM)
GPIO 25 → IN3	Right motor forward		
GPIO 33 → IN4	Right motor backward		

IMPORTANT: Remove ENA and ENB jumpers from the L298N board before wiring.

Network settings

WiFi SSID : ESP32-RC-Car
WiFi Password: rccar123 (minimum 8 characters, WPA2)

Controller URL: <http://192.168.4.1> (always works)
or <http://esp32car.local> (most phones support this)

Fail-safe timeout: 400 ms (motors stop if no command received)
PWM frequency: 5000 Hz (above audible whine range)
PWM resolution: 8-bit (0-255 speed range)

Libraries to install

1. ESPAsyncWebServer — author: ESP32Async
<https://github.com/ESP32Async/ESPAsyncWebServer>
 2. AsyncTCP — author: ESP32Async
<https://github.com/ESP32Async/AsyncTCP>
- Board package: "esp32 by Espressif Systems" v3.x
URL: https://dl.espressif.com/dl/package_esp32_index.json

ESP32 WiFi RC Car — Build Walkthrough — All diagrams and content original